Microeconomics

The consumer problem

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Artificial Intelligence and Financial Technologies
New Economic School · 2026/27

From preferences to an actual choice

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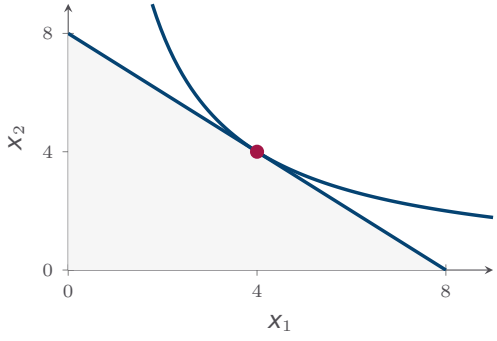
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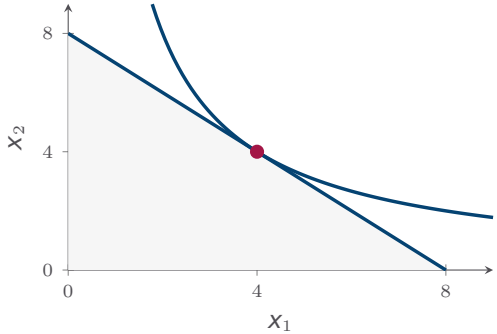
Expenditure minimisation, Hicksian demand and Slutsky belong to Lecture 3.

The picture we already know



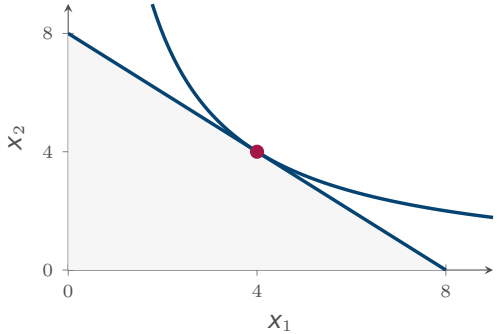
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$u(x) = \sqrt{x_1 x_2}$, $p = (1, 1)$, $m = 8$. Which affordable bundle is best? $(4, 4)$. Today we want a systematic method, not just a convincing picture.

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Preferences enter through u; prices and wealth enter through the feasible set.

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What changes for a subscription, a quantity discount or an indivisible purchase?

Two useful facts about the budget

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Nonemptiness: the zero bundle is affordable.
Compactness: the budget is closed, and $0 \leq x_i \leq m/p_i$ for every i.
Strictly positive prices give a finite upper bound on every quantity.

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In finite-dimensional Euclidean space, closed and bounded means compact.

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Existence does not yet imply uniqueness, an interior solution or differentiability of demand.

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Can you give a case with multiple maximisers even though the budget is compact?

For $u = x_1 + x_2$, $p = (1, 1)$ and $m = 1$, every bundle on $x_1 + x_2 = 1$ is optimal.

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It is the earlier choice rule applied specifically to budget menus:
$D(p, m) = C(B(p, m))$.

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Example: $(p, m) = ((2, 1), 12)$ and $((200, 100), 1200)$.

Budget exhaustion: Walras' law

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Assume an optimum exists. What would happen if it left strictly positive budget slack?

The key: with slack, every sufficiently small change in the consumption domain remains affordable.

Proof: LNS and budget exhaustion

Suppose an optimum x^* has slack $\delta = m - p \cdot x^* > 0$.

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If $\|y - x^*\| < \delta / \|p\|$, then

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LNS supplies a strictly better such y in the consumption domain, contradicting optimality.

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Convex preferences imply $z_t \succeq w$ for every such w. Thus $z_t \in D(p, m)$.
Hence $D(p, m)$ is convex. It can still be a whole line segment.

Strict convexity: at most one optimum

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No. Continuity and compactness gave existence; strict convexity gives at most one optimum.

Proof: uniqueness from strict convexity

Suppose distinct $x, y \in D(p, m)$. Their midpoint z is feasible because the budget is convex.

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This contradicts x being optimal. There is at most one optimum; existence gives a unique one.

→ Back to uniqueness

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Demand tells us **which bundle**; indirect utility tells us **the attained value**. Even if $D(p, m)$ contains several bundles, every one gives the same value $v(p, m)$. It is a function of the environment, not of a chosen bundle.

Indirect utility: the properties to prove

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v is continuous on the positive-price, positive-wealth domain. No differentiability claim yet.

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Thus $q \cdot x \leq m$ implies $p \cdot x \leq m$: $B(q, m) \subseteq B(p, m)$.
Maximising the same utility over a smaller set cannot give a larger value: $v(q, m) \leq v(p, m)$.

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$p \cdot y \leq p \cdot x + \|p\|\|y - x\| < m + (m' - m)/2 < m'$.
So y is affordable at the new wealth and $v(p, m') \geq u(y) > v(p, m)$.

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This is a bijection between $B(p, m)$ and Δ, since $p \cdot x / m = \sum_i z_i$.
With $g(p, m, z) = u((m/p_1)z_1, \ldots, (m/p_L)z_L)$, we have $v(p, m) = \max_{z \in \Delta} g(p, m, z)$.

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g is continuous on the compact set $K \times \Delta$, hence uniformly continuous.

Therefore $\sup_{z \in \Delta} |g(p^n, m^n, z) - g(p, m, z)| \to 0$.

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$$|v(p^n, m^n) - v(p, m)| \leq \sup_{z \in \Delta} |g(p^n, m^n, z) - g(p, m, z)| \to 0.$$

No differentiability or uniqueness of the optimiser was needed.

A less obvious property: quasiconvexity

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So the mixed budget cannot deliver utility above both endpoint values. → Proof

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Thus every bundle affordable at the mixture is affordable at at least one endpoint.
Its utility cannot exceed $\max\{v(p, m), v(q, n)\}$. Maximise over the mixed budget.

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Complements: look for useful proportions; the optimum may be a kink.

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Complements: look for useful proportions; the optimum may be a kink.

Smooth interior problem: marginal conditions can identify a candidate.

Always check feasibility, boundary cases and a reason why the candidate is globally optimal.

Let a multiplier enforce the budget

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One variable chooses consumption; the other enforces the budget. Their directions of optimisation differ.

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Thus the original problem can be written $\max_{x \geq 0} \inf_{\lambda \geq 0} \ell(x, \lambda)$.
This identity does not authorise exchanging max and inf.

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Left inequality: x^* maximises ℓ over the consumption domain, holding λ^* fixed.
Right inequality: λ^* minimises ℓ over nonnegative multipliers, holding x^* fixed.

Proof: the saddle enforces feasibility and slackness

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Set $\lambda = 0$: $\lambda^* s \leq 0$. Nonnegative λ^* and s also give $\lambda^* s \geq 0$.

Therefore $\lambda^*(m - p \cdot x^*) = 0$ and $\ell(x^*, \lambda^*) = u(x^*)$.

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This direction needs no concavity or differentiability: the saddle inequalities are already global.

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We now make the bounds $x_i \geq 0$ explicit using multipliers μ_i. → KKT back to a saddle

To the board: derive Kuhn–Tucker

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Start from the budget and the nonnegativity bounds.
Derive stationarity, primal and dual feasibility, and complementary slackness.
Then separate necessity from global sufficiency.

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If both goods are consumed in a smooth optimum, these marginal returns must agree.

The Lagrangian: one multiplier per inequality

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For the smooth argument, assume u is differentiable on an open neighbourhood of the candidate.

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Stationarity is only one group of conditions. We must also check signs, feasibility and slackness.

The four Kuhn–Tucker groups

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Primal feasibility: $p \cdot x^* \leq m$, $x^* \geq 0.$
Dual feasibility: $\lambda^* \geq 0$, $\mu^* \geq 0.$
Complementary slackness:

$$\lambda^*(m - p \cdot x^*) = 0, \qquad \mu_i^* x_i^* = 0.$$

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Complementary slackness: read the products

If the budget is slack, $m - p \cdot x^* > 0$, then $\lambda^* = 0$.
If $x_i^* > 0$, then $\mu_i^* = 0$.
If a multiplier is strictly positive, its constraint must bind.
Does a binding constraint always have a strictly positive multiplier?

For consumed goods: equal marginal returns

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This is an interior condition, not a universal rule for every optimum.

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If $x_i^* = 0$, stationarity and $\mu_i^* \geq 0$ imply

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Why would a rule that sets every marginal return equal miss some solutions?

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Necessity: a differentiable local optimum satisfies KKT under a constraint qualification.

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Quasiconcavity alone does not make every stationary point optimal. → Sufficiency proof → Counterexample

Checkpoint

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When is a KKT solution sufficient for global optimality?
What can go wrong if we check only the interior first-order equations?

Example 1: Cobb–Douglas

$$u(x) = x_1^{\alpha} x_2^{1-\alpha}, \qquad 0 < \alpha < 1.$$

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Therefore every optimum is strictly positive. The nonnegativity multipliers are zero.

Now an interior method is justified; it was not assumed at the start.

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The bundle does not change, but the numerical utility and multiplier generally do.

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$$x_1(p, m) = \frac{\alpha m}{p_1}, \qquad x_2(p, m) = \frac{(1-\alpha)m}{p_2}.$$

Interpret the solution

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If $\alpha = 1/2$, $p = (2, 1)$ and $m = 12$, demand is $(3, 6)$.

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If $\alpha = 1/2$, $p = (2, 1)$ and $m = 12$, demand is $(3, 6)$.

Which of these predictions is a modelling restriction rather than an accounting identity?

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Does an unchanged quantity of good 2 mean that its utility contribution is unchanged?

Example 2: perfect substitutes

$$u(x) = ax_1 + bx_2, \qquad a, b > 0.$$

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If $a/p_1 > b/p_2$, buy $(m/p_1, 0)$. If the inequality reverses, buy $(0, m/p_2)$.
If the ratios are equal, every point on the budget line is optimal.

Verify a corner with the full KKT system

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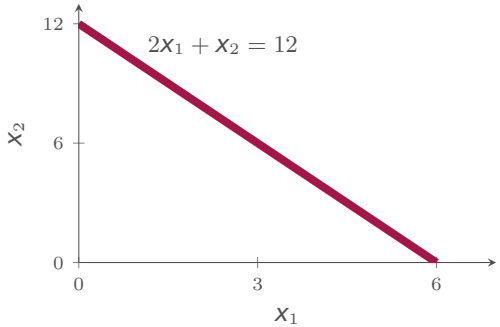
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Check: $2 - 3\lambda + \mu_1 = 0$, $1 - \lambda + \mu_2 = 0$.

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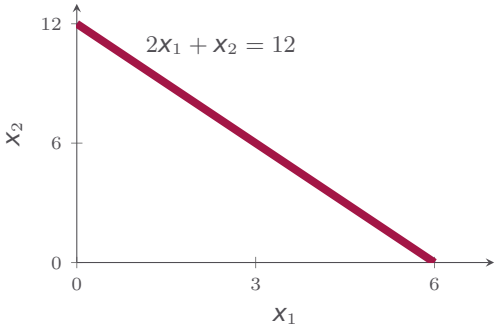
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Check: $2 - 3\lambda + \mu_1 = 0$, $1 - \lambda + \mu_2 = 0$.
The zero quantity is optimal precisely because its marginal return is too low.

Equal returns: a whole set of demand



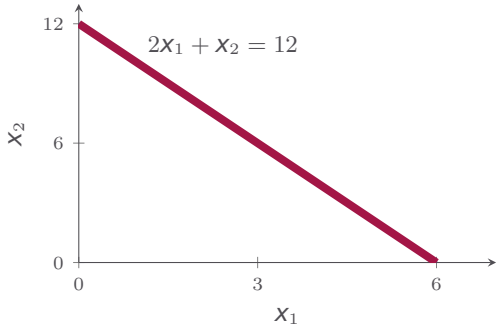
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Every budget-line bundle gives utility 12.

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$u = 2x_1 + x_2$, $p = (2, 1)$, $m = 12$.
Every budget-line bundle gives utility 12.
$D(p, m)$ is a segment, while $v(p, m) = 12$ is a single number.

Example 3: perfect complements

$$u(x) = \min\{x_1, x_2\}.$$

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The optimum is a kink: the smooth stationarity formula does not apply there.

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If $0 < m < 4$: $a^* = 0$, $\lambda = 2/\sqrt{m} > 1$, $\mu_a = \lambda - 1 > 0$.

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Fitting one choice does not establish the functional form. Use more variation and test the restrictions.

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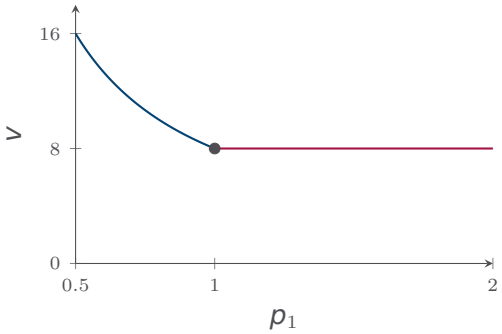
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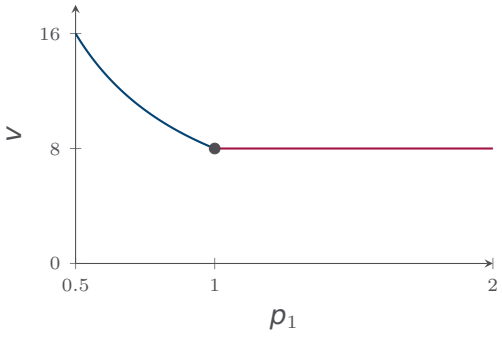
$\rightarrow$ More than two goods

A continuous value need not be differentiable



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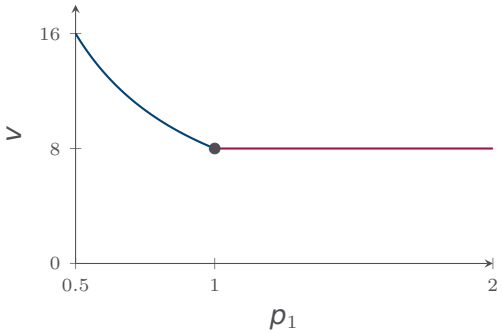
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At $p_1 = 1$, the consumer is indifferent along the budget line. The value has a kink.

The envelope idea

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We need appropriate regularity and a differentiable value; a kink is a warning.

Differentiate the value, not the choice

At a regular optimum where the envelope formulas apply,

$$v_m(p, m) = \mathcal{L}_m = \lambda^*.$$

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The wealth effect is positive when $\lambda^* > 0$; a higher price hurts in proportion to the quantity purchased.

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A binding budget does not by itself imply $v_m > 0$. → A counterexample

Roy's identity: recover demand from value

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Under these conditions the optimal quantity vector is unique. → Derivation

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Likewise $x_2 = (1-\alpha)m/p_2$. The demand agrees with the direct solution.

Demand survives a change of utility scale

If $\widehat{v} = f(v)$ with $f'(v) > 0$, then

$$\widehat{v}_{p_i} = f'(v) v_{p_i}, \qquad \widehat{v}_m = f'(v) v_m.$$

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A strictly increasing transformation can have a zero derivative at a point; then the differentiable ratio may be unavailable.

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What two differentiability/positivity conditions are needed to use Roy's formula?

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Homework 1 covers Lectures 1–2 and is due on 14 November 2026.

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One fixes resources; the other fixes a level of the same preferences.

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A continuous cost attains a minimum there; excluded bundles are more expensive. → Details

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No excess utility uses continuity and positive prices, not LNS.

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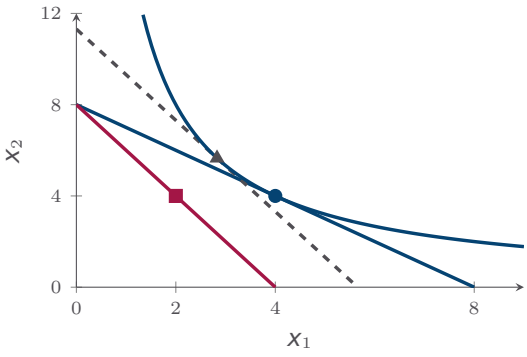
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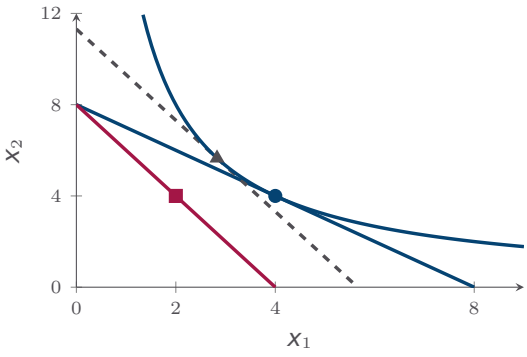
It is the infimum of linear cost functions. → Proofs

Fixed wealth versus fixed utility



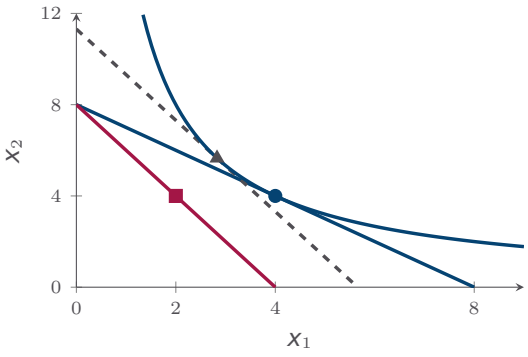
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To keep utility 4, spend $8\sqrt{2}$ and buy $(2\sqrt{2}, 4\sqrt{2})$.

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These connect the two problems; they are not an assumption that the consumer literally solves both.

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A money-metric change is useful, but summing it across people still involves welfare judgements.

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→ Back to necessity and sufficiency

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The active gradients are therefore linearly independent. This includes corner optima.

→ Back to necessity and sufficiency

Proof: why concavity makes KKT sufficient

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Stationarity gives $\nabla u(x^*) = \lambda^* p - \mu^*$.
Budget feasibility and slackness give $\lambda^* p \cdot (y - x^*) \leq 0$. Bound feasibility and slackness give $-\mu^* \cdot (y - x^*) \leq 0$.

→ Back

Proof: why concavity makes KKT sufficient

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Thus $u(y) \leq u(x^*)$ for every feasible y.

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Proof: concave KKT gives the budget saddle

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Both saddle inequalities hold. → Back to the bridge

A KKT point that is not optimal

Let $u(x) = (x-1)^3$, $p = 1$, $m = 2$, $x \geq 0$.

→ Back

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NES

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The function is strictly increasing and quasiconcave. That alone does not make stationarity sufficient.

→ Back

A binding budget with zero marginal value

Use $u(x) = (x-1)^3$, but now $p = 1$, $m = 1$.

→ Back to the multiplier

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Roy's ratio becomes $0/0$ at that point, although demand $x = m/p$ is perfectly well-defined.

→ Back to the multiplier

Cobb–Douglas with any positive exponent sum

Let $u(x) = \prod_{i=1}^{L} x_i^{a_i}$ with $a_i > 0$ and $A = \sum_i a_i$.

→ Back

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No restriction $A \leq 1$ is needed for this preference solution.

→ Back

Roy without differentiating demand

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→ Back

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If $v_m > 0$, every optimiser satisfies the same Roy formula, so demand is unique.

→ Back

Proof: expenditure minimisation has a solution

Choose an attainable target and a feasible $\widehat{x}$. Let

$$S = \{x \geq 0 : u(x) \geq \bar{u}\} \cap \{x : p \cdot x \leq p \cdot \widehat{x}\}.$$

→ Back to the preview

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S is nonempty and compact when u is continuous and $p \gg 0$.
Cost is continuous, so it has a minimum on S. Every omitted feasible bundle costs more than $\widehat{x}$, so this is a global minimum.

→ Back to the preview

Proof: homogeneity and price concavity of expenditure

Scaling prices multiplies every feasible cost by the same positive number:

$$e(tp, \bar{u}) = te(p, \bar{u}), \qquad H(tp, \bar{u}) = H(p, \bar{u}).$$

→ Back to expenditure properties

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$$e(p^t, \bar{u}) = tp \cdot x^t + (1-t)q \cdot x^t \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

→ Back to expenditure properties